

X100/303

NATIONAL
QUALIFICATIONS
2007

TUESDAY, 15 MAY
10.30 AM – 12.00 NOON

MATHEMATICS
HIGHER
Units 1, 2 and 3
Paper 2

Read Carefully

- 1 **Calculators may be used in this paper.**
- 2 Full credit will be given only where the solution contains appropriate working.
- 3 Answers obtained by readings from scale drawings will not receive any credit.



FORMULAE LIST

Circle:

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r .

Scalar Product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae: $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2\cos^2 A - 1$$

$$= 1 - 2\sin^2 A$$

Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

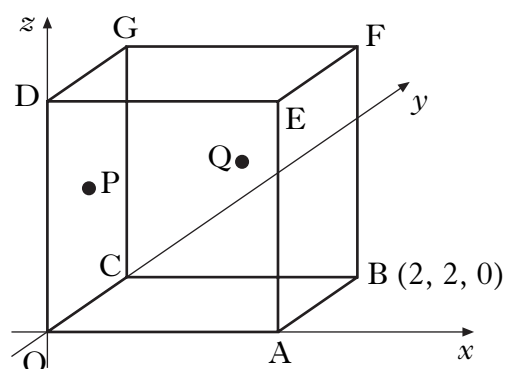
Table of standard integrals:

$f(x)$	$\int f(x) dx$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

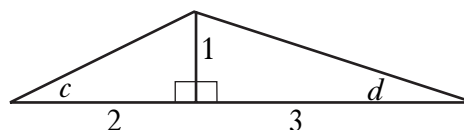
1. OABCDEFG is a cube with side 2 units, as shown in the diagram.

B has coordinates $(2, 2, 0)$.

P is the centre of face OCGD and Q is the centre of face CBF G.

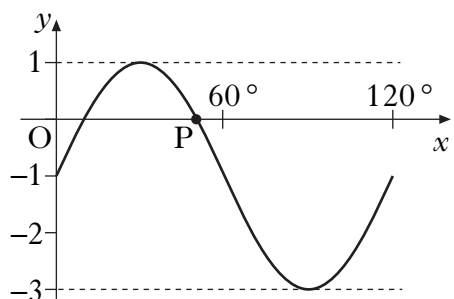


- (a) Write down the coordinates of G.
- (b) Find $\mathbf{p}$ and $\mathbf{q}$, the position vectors of points P and Q.
- (c) Find the size of angle POQ.
2. The diagram shows two right-angled triangles with angles c and d marked as shown.



- (a) Find the exact value of $\sin(c + d)$.
- (b) (i) Find the exact value of $\sin 2c$.
- (ii) Show that $\cos 2d$ has the same exact value.
3. Show that the line with equation $y = 6 - 2x$ is a tangent to the circle with equation $x^2 + y^2 + 6x - 4y - 7 = 0$ and find the coordinates of the point of contact of the tangent and the circle.

4. The diagram shows part of the graph of a function whose equation is of the form $y = a \sin(bx^\circ) + c$.

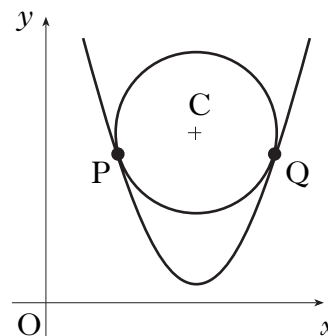


- (a) Write down the values of a , b and c .
- (b) Determine the exact value of the x -coordinate of P, the point where the graph intersects the x -axis as shown in the diagram.

[Turn over

5. A circle centre C is situated so that it touches the parabola with equation $y = \frac{1}{2}x^2 - 8x + 34$ at P and Q .

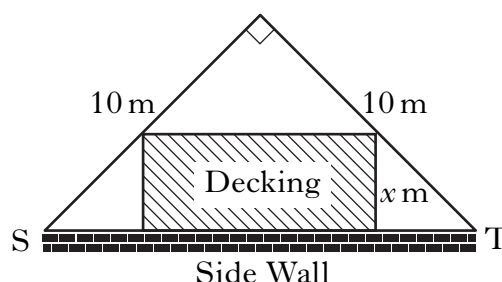
- (a) The gradient of the tangent to the parabola at Q is 4. Find the coordinates of Q .
- (b) Find the coordinates of P .
- (c) Find the coordinates of C , the centre of the circle.



5
2
2

6. A householder has a garden in the shape of a right-angled isosceles triangle.

It is intended to put down a section of rectangular wooden decking at the side of the house, as shown in the diagram.



- (a) (i) Find the exact value of ST .
- (ii) Given that the breadth of the decking is x metres, show that the area of the decking, A square metres, is given by

$$A = (10\sqrt{2})x - 2x^2. \quad 3$$

- (b) Find the dimensions of the decking which maximises its area. 5

7. Find the value of $\int_0^2 \sin(4x+1) dx$. 4

8. The curve with equation $y = \log_3(x-1) - 2 \cdot 2$, where $x > 1$, cuts the x -axis at the point $(a, 0)$.

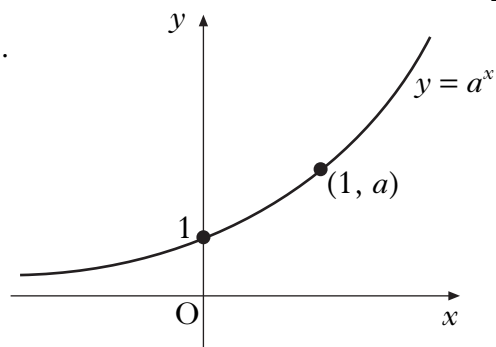
Find the value of a . 4

9. The diagram shows the graph of $y = a^x$, $a > 1$.

On separate diagrams, sketch the graphs of:

(a) $y = a^{-x}$;

(b) $y = a^{1-x}$.



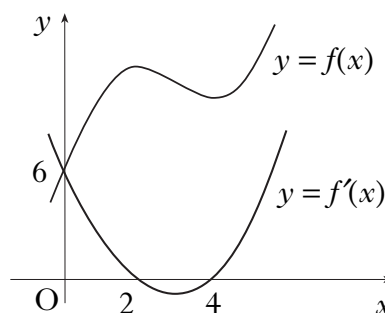
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2

10. The diagram shows the graphs of a cubic function $y = f(x)$ and its derived function $y = f'(x)$.

Both graphs pass through the point $(0, 6)$.

The graph of $y = f'(x)$ also passes through the points $(2, 0)$ and $(4, 0)$.



- (a) Given that $f'(x)$ is of the form $k(x - a)(x - b)$:

(i) write down the values of a and b ;

(ii) find the value of k .

3

- (b) Find the equation of the graph of the cubic function $y = f(x)$.

4

11. Two variables x and y satisfy the equation $y = 3 \times 4^x$.

(a) Find the value of a if $(a, 6)$ lies on the graph with equation $y = 3 \times 4^x$.

1

(b) If $(-\frac{1}{2}, b)$ also lies on the graph, find b .

1

(c) A graph is drawn of $\log_{10}y$ against x . Show that its equation will be of the form $\log_{10}y = Px + Q$ and state the gradient of this line.

4

[END OF QUESTION PAPER]